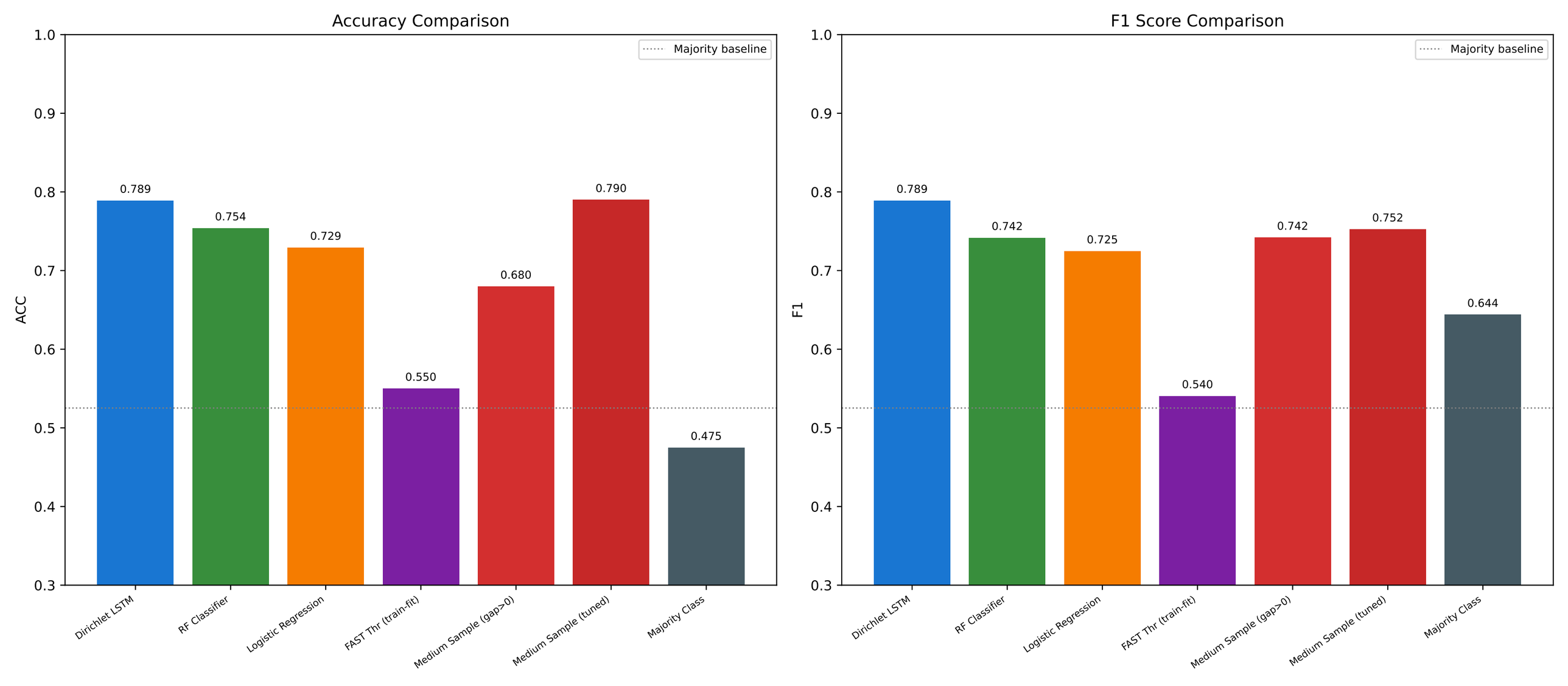
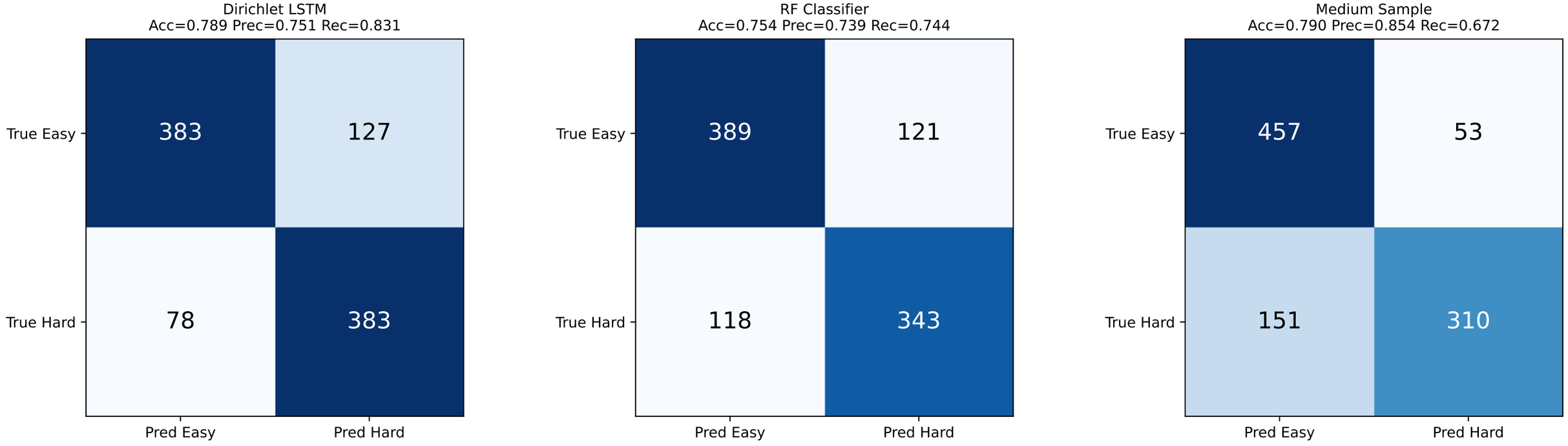
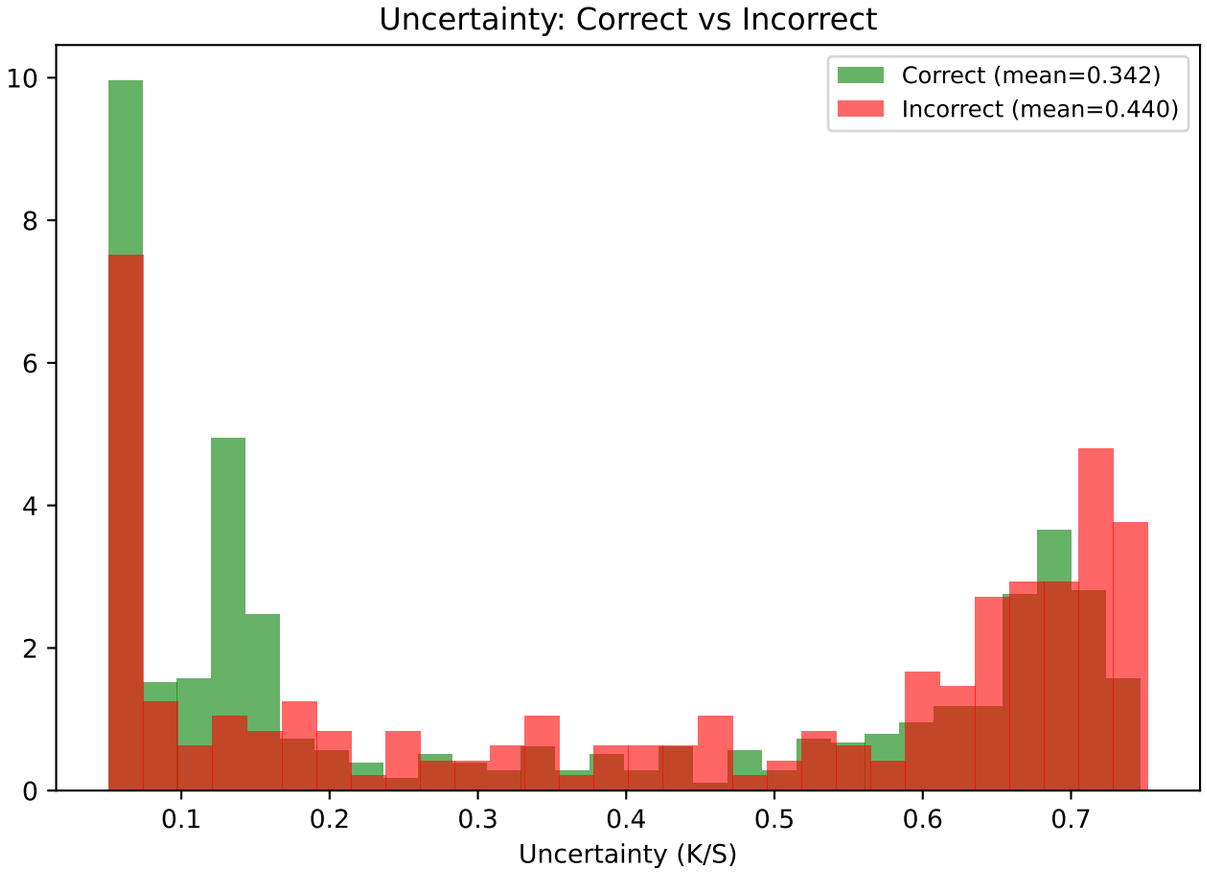
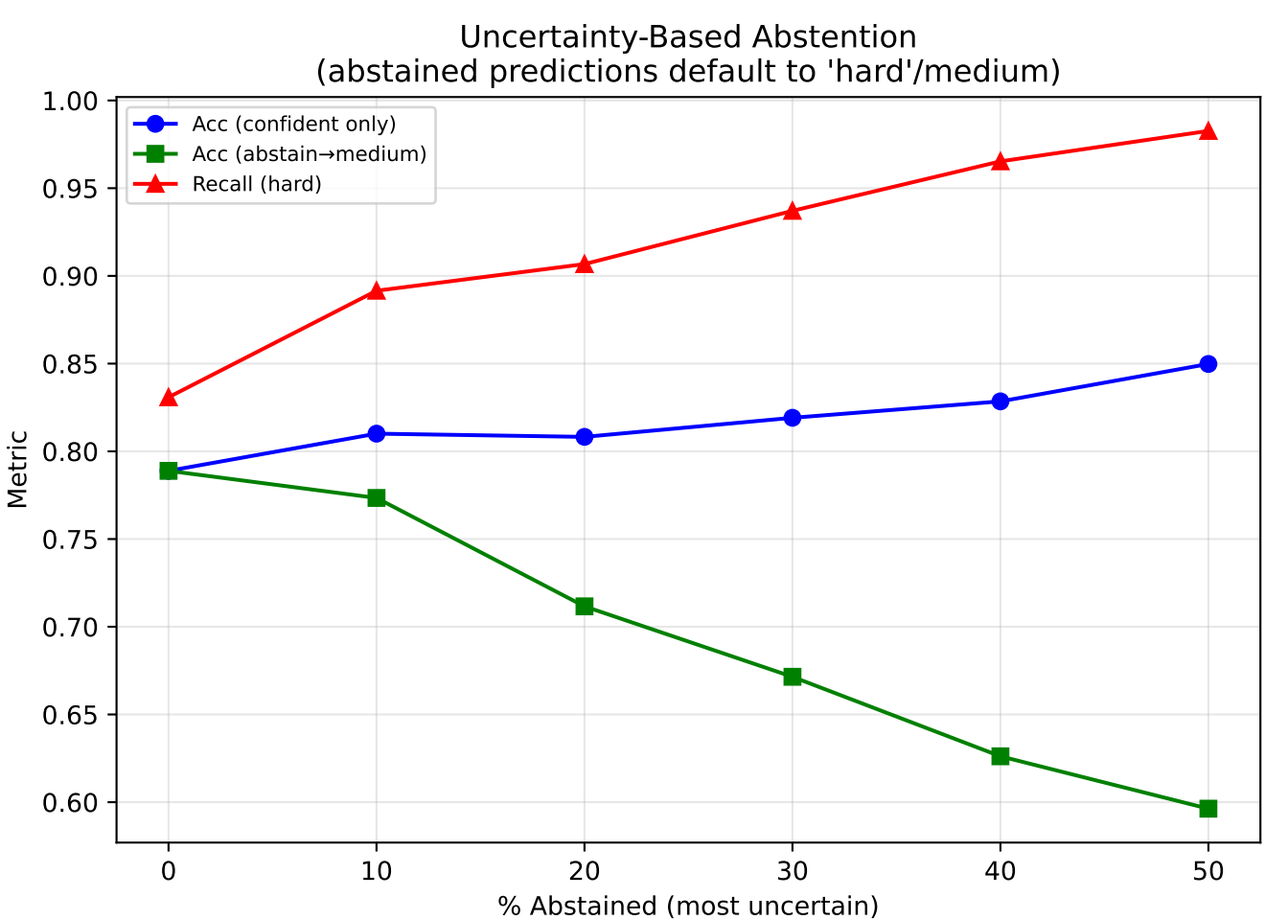
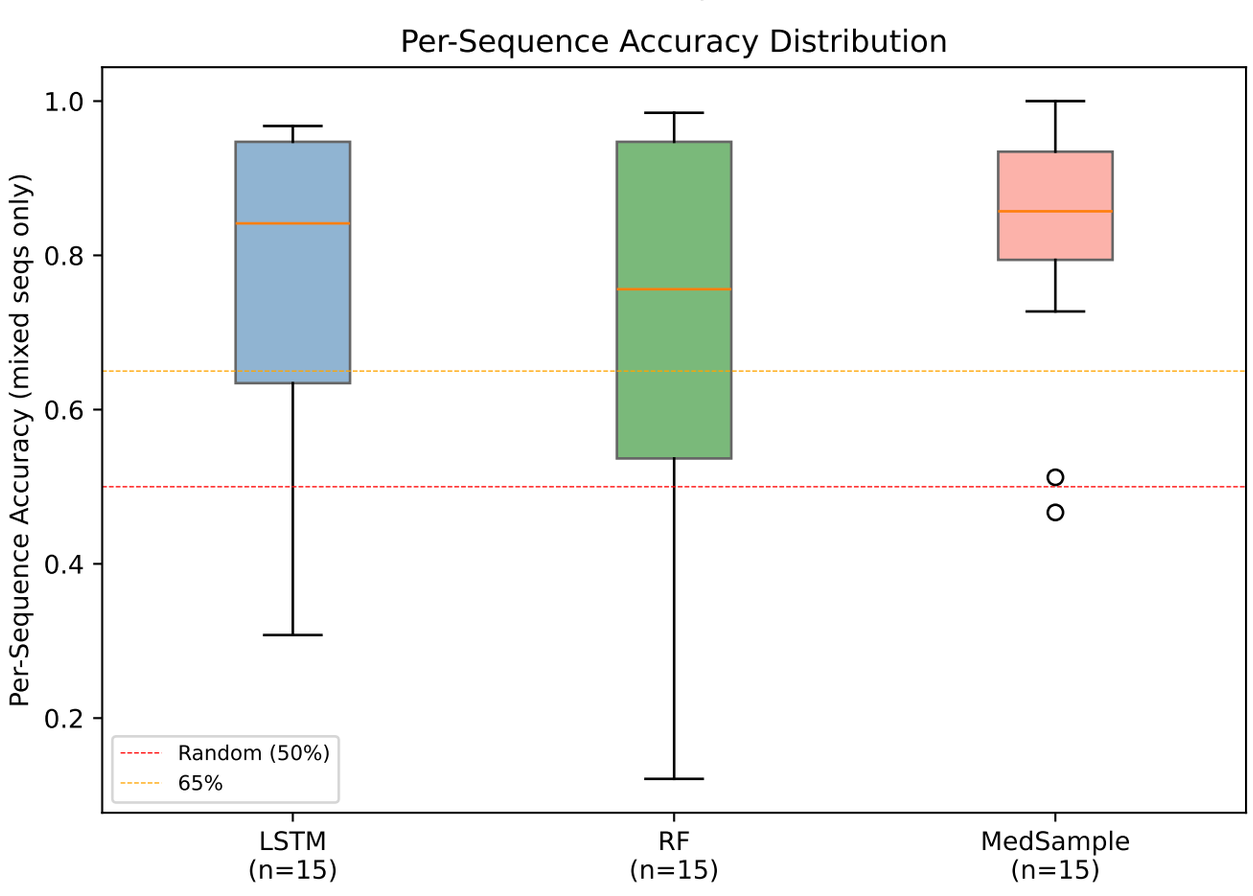
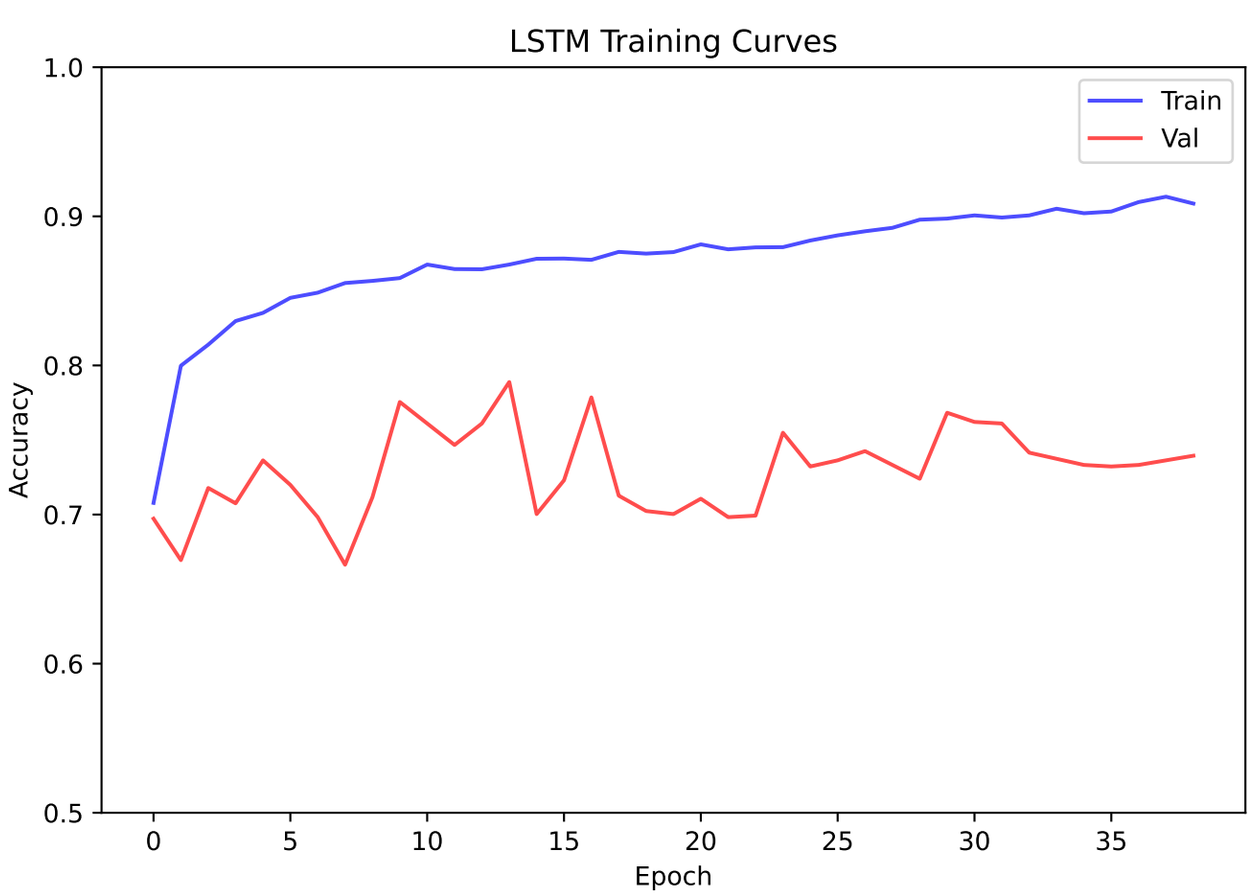

Accuracy Comparison
ACC
Majority baseline
0.789
0.754
0.729
0.550
0.680
0.790
0.475
Dirichlet LSTM
RF Classifier
Logistic Regression
FAST Thr (train-fit)
Medium Sample (gap>0)
Medium Sample (tuned)
Majority Class
F1 Score Comparison
F1
Majority baseline
0.789
0.742
0.725
0.540
0.742
0.752
0.644
Dirichlet LSTM
RF Classifier
Logistic Regression
FAST Thr (train-fit)
Medium Sample (gap>0)
Medium Sample (tuned)
Majority Class

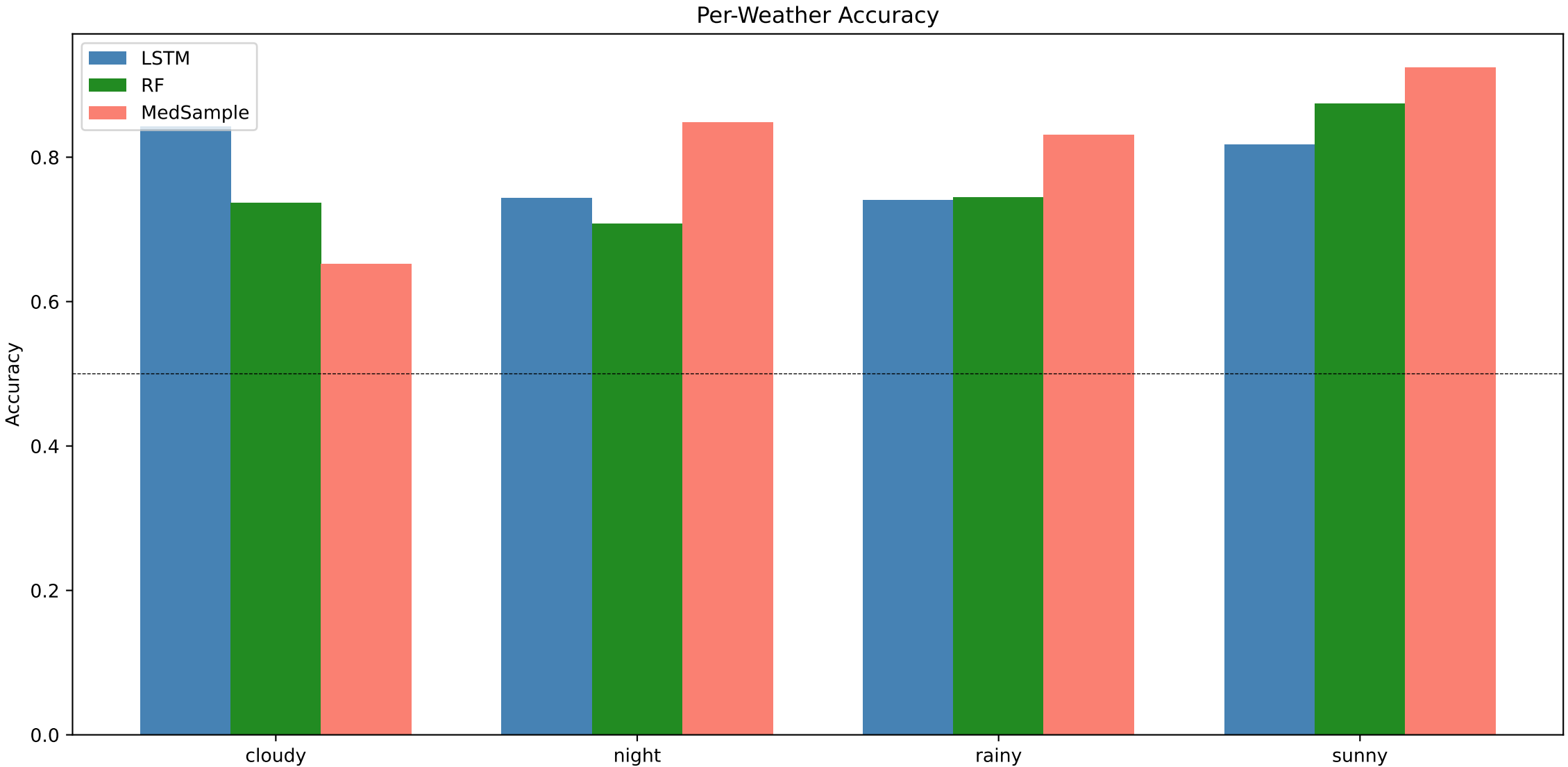

Confusion Matrices — Who Gets It Wrong and How?
Dirichlet LSTM
Acc=0.789 Prec=0.751 Rec=0.831
True Easy
True Hard
383
127
78
383
Pred Easy
Pred Hard
RF Classifier
Acc=0.754 Prec=0.739 Rec=0.744
True Easy
True Hard
389
121
118
343
Pred Easy
Pred Hard
Medium Sample
Acc=0.790 Prec=0.854 Rec=0.672
True Easy
True Hard
457
53
151
310
Pred Easy
Pred Hard


Uncertainty: Correct vs Incorrect
Correct (mean=0.342)
Incorrect (mean=0.440)
Uncertainty (K/S)


Uncertainty-Based Abstention
(abstained predictions default to 'hard'/medium)
Acc (confident only)
Acc (abstain→medium)
Recall (hard)
Metric
% Abstained (most uncertain)


Per-Sequence Accuracy Distribution
Per-Sequence Accuracy (mixed seqs only)
Random (50%)
65%
LSTM
(n=15)
RF
(n=15)
MedSample
(n=15)


LSTM Training Curves
Train
Val
Accuracy
Epoch


Per-Weather Accuracy
LSTM
RF
MedSample
Accuracy
0.0
0.2
0.4
0.6
0.8
cloudy
night
rainy
sunny

Complete Evaluation — Dirichlet LSTM vs All Baselines

Model	Acc	F1	Prec	Rec	AUC	Deployable?	Cost
Dirichlet LSTM	0.789	0.789	0.751	0.831	0.848	Yes	20 features/frame
RF Classifier	0.754	0.742	0.739	0.744	0.824	Yes	20 features/frame
Logistic Regression	0.729	0.725	0.700	0.751	0.809	Yes	20 features/frame
FAST Thr (train-fit)	0.550	0.540	0.524	0.557	—	Yes	1 feature/frame
Medium Sample (gap>0)	0.680	0.742	0.601	0.970	—	Yes	1 medium/sec (3.3%)
Medium Sample (tuned)	0.790	0.752	0.854	0.672	—	Yes	1 medium/sec (3.3%)
Majority Class	0.475	0.644	0.475	1.000	—	Yes (trivial)	Zero

Task: predict whether mean(fn_nano) over next 30 frames >= 2.70
Class balance: 47.5% hard / 52.5% easy (majority baseline: 52.5%)

Key findings:
1. Dirichlet LSTM (78.9%) and RF (75.4%) have COMPARABLE accuracy.
LSTM's advantage is calibrated uncertainty, not prediction quality.
2. Medium-sampling baseline (79.0%): costs 1 medium inference/sec.
BEATS Dirichlet LSTM.
3. Dirichlet uncertainty IS calibrated (incorrect: 0.440 > correct: 0.342).
4. Abstaining on 20% most uncertain + defaulting to medium → recall=0.907